The answer keys against the literature, and what they can decide

Written 2026-10-16 against the build of 2026-10-13 (2,452 entries), and **updated 2026-10-31 to the build of 2026-10-16 (3,291 entries)**, after D25–D27 entered 841 molecules because a public list names them. Every figure in §2–§4 was measured again on the larger build. §6 is new: what adding those molecules did to the keys, and what adding more would do. The 2,452-entry figures appear only where §6 sets the two builds side by side. **A review and a proposal; nothing in it is adopted, and no key, constant or page behaviour was changed.** Every measurement is reproduced by a read-only script under `analysis/answer_keys/`, named where it is used.

SIMILARITY-LITERATURE.md (2026-10-13) asked what the literature says about the *score*, and proposed four better keys. All four were built by 2026-10-14 (SIMILARITY-HARNESS.md §13–16; explained in ANSWER-KEYS-PRIMER.md). This document turns the same question on the *keys*. They are about to be used to decide changes to the score, so it asks: what does the literature on reference standards say about keys of each kind, and can these keys actually decide the changes they will be asked to decide?

In short:

- **The keys cannot see the changes the sweep reports.** Compared paired, question by question, the smallest change a key can detect is 0.010 MRR (MeSH) to 0.019 (repaired ATC). The changes §15 of the harness sorts into "agreed by every judge", "agreed only by the curated judges" and "split" are mostly +0.002 to +0.009. Under a paired test with Holm's correction across the 42 key-by-setting cells, **only switching D15 off is detected under every key.** Raising the family weight to 1.0 is detected under MeSH, EPC and PE, and under neither label reading (on the 2,452-entry build, under MeSH alone). The other four settings are within every key's noise, including the curated-only gain and the replicated-only split that §15 reads as a finding. To detect a +0.004 change at 80% power would need 3,200 to 6,400 questions depending on the key; the most any key asks is 978 (§2).
- **The label key is noisier than the ranking it grades.** Build it twice, once from US labels and once from UK, EU and Japanese labels, and the two share 14% of their five nearest neighbours. The target ranking predicts each half's neighbours *better* than the other half does (§3.4). This fits the literature: a US and an EU label for the same CNS drug share 29% of their adverse-reaction terms (Cornelius 2016).
- **The label key's agreement with the ranking runs through shared indication.** Label neighbours share an approved indication 41% of the time, random pairs 8%. Rebuilt so that a neighbour must be approved for a different disease, the key agrees with the ranking at top-1 1.6% against a shuffle's 0.7% (§3.4). It is independent of the score's inputs, but not of what the drugs are for.
- **Most of the pairwise key's negatives cannot fail.** 96 of the 160 negative directions share no target, so they are absent from both lists and clean by construction. 110 of the 116 positives are already classmates under the repaired ATC, EPC or MeSH key (§3.5). Its value is the opposites (all 94 directions reachable) and the 64 negative directions that could fail.

- **The class keys are incomplete by construction, which the literature measures.** FDA's asserted class memberships match those inferable from its own drug descriptions about half the time (Winnenburg 2015), and ATC is extended only on request (Liu 2015). A miss under a class key is often a missing label, which supports reading the filtered grade first (§3.1–3.3).
- **A larger catalogue does not make the keys sharper.** A key gains power only from questions, and a question needs a label. The 841 molecules D25–D27 entered are 25.6% of the catalogue. They are no ATC question, 112 MeSH questions, 11 EPC, 15 PE and 4 label-key questions. The headline ATC key cannot gain a question at all: every ATC N molecule in ChEMBL 37 is already in. What the new molecules did do is crowd every list. Grades as built fell (ATC MRR 0.4607 to 0.4341), and so did the pairwise key's positives within the top 10 (75.4% to 69.8%), but the ranking among the old molecules did not get worse: read with the listed entries skipped, the same build grades 0.4644 and 75.0%. The label key's answers moved too, and seven of the ten molecules that appear most often in others' top 10 are now listed entries. Even entering every approved drug in ChEMBL 37, most of them outside the brain scope, would bring the best key to about 0.007, not 0.004 (§6).
- **Proposed order (§7):** fix the decision rule before the next constant is moved (no new data); a pairwise key v2 with negatives that can fail and a second, independent drafter; a label key v2 that requires agreement across regulators and removes a drug's own indication symptoms; then pilots of brain-specific keys. None of the brain-specific keys the literature offers is ready to build at catalogue scale (§5).

---

1. What a key has to do now

SIMILARITY-LITERATURE §4.1 listed six properties for a key: it asks the page's question, states its provenance, judges only what it can judge, grades with negatives, comes as several judges with one held out, and is fixed before use. The keys as built meet most of them. Using them to decide changes to the score adds four more, each from a literature that has faced the same problem:

property	why	from
It resolves the effect sizes at issue. A grade's sampling noise must be smaller than the change being judged, tested on paired per-question differences, with the number of settings tried counted.	Information retrieval compares two systems over one question set the same way, and found that many reported gains do not survive a paired test.	Smucker, Allan & Carterette 2007; Sakai 2016; Carterette 2012
Its own reliability is known. A key that disagrees with a second measurement of itself caps the grade any ranking can reach.	The "noise ceiling" of representational similarity analysis.	Nili et al. 2014
Its negatives can fail. A negative control the method cannot get wrong adds nothing to a rate but its denominator, and negatives are where reference sets are most often misclassified.	Drug-safety reference sets with positive and negative controls.	Ryan et al. 2013; Coloma et al. 2013; Hauben et al. 2016

property	why	from
What it shares with the predictor is measured, not only stated.	Circularity inflated the evaluation of variant-effect predictors, and redundancy between benchmark and training data explained most of the performance of ligand-based classifiers.	Grimm et al. 2015; Wallach & Heifets 2018

The rest of this document checks each key against these properties.

---

2. Can the keys see the changes? (`power.mjs`)

`node analysis/answer_keys/power.mjs` takes the six settings in SIMILARITY-HARNESS §15 and, for each key, compares every question's reciprocal rank before and after. It reports a paired bootstrap interval on the MRR change. For the class keys it adds a second bootstrap that resamples whole classes, since one constant moves a class's members together. Then come a sign-flip randomisation test, and Holm's correction across all 42 cells. A cell counts below as **detected** when it survives Holm.

How precisely each grade is known. These are baseline MRRs with a 95% bootstrap interval over questions:

key	questions	MRR	95% interval
ATC (headline)	284	0.4341	0.388 – 0.481
ATC repaired	259	0.4640	0.414 – 0.514
MeSH	978	0.5607	0.535 – 0.586
EPC	469	0.5650	0.528 – 0.601
PE	612	0.4984	0.465 – 0.530
label, nervous system	350	0.2512	0.217 – 0.287
label, every effect	514	0.3056	0.274 – 0.338

What each setting does, under seven keys. The full table, with intervals and the count of questions moved each way, is the script's output.

setting	MRR change across the seven keys	keys whose interval excludes zero	detected (Holm)
D15 off (`OPPOSED_PAIR_FACTOR` 1.0)	−0.015 to −0.029	7	**7**
family weight 0.5 → 1.0	+0.003 to +0.019	4 (MeSH, EPC, PE, label every effect)	3 (MeSH, EPC, PE)
curated-only 0.75 → 1.0	+0.004 to +0.008	1 (MeSH)	0
replicated-only 0.9 → 0.75	−0.002 to +0.007	0	0
Preliminary 0.5 → 0.25	−0.018 to +0.009	0	0
`SPECIES_EXPONENT` 0.5 → 0.75	−0.000 to +0.005	0	0

Resampling whole classes widens most class-key intervals, by up to 98% for D15 under the repaired ATC key, since one change moves a class together. Of the 30 class-key cells, 9 still exclude zero under it, against 12 of all 42 cells at the question level.

What this changes in §15's reading. §15 sorted the constants into three kinds by the sign of their change under five judges:

- **"Agreed by every judge."** D15 holds: it is the one finding every key resolves. The other three are not shown by these keys to do anything, though §15 reads them as small agreed gains. The curated-only gain is the case that matters, because §15 and CLAUDE.md read it as answering the old caveat that the judge flattered curation. Under the label key it is +0.0045, interval −0.004 to +0.013.
- **The family weight** is the one setting near the line, and the line moved under it. On this build MeSH, EPC and PE detect it, though only MeSH did on the 2,452-entry build. With the listed entries skipped in each list, MeSH and PE detect it and EPC does not (`power.mjs --skip-listed`). The keys did not get sharper here. The catalogue changed what the setting does (§6). No key that reads no curation detects it: the every-effect label reading's interval excludes zero, but the cell does not survive Holm's correction.
- **"Agreed only by the curated judges."** The replicated-only change helps under ATC and MeSH and "hurts" under both label readings, and §15 calls this "the pattern the label key was built to catch". The label key's intervals are −0.009 to +0.006 and −0.009 to +0.004. It has not caught anything yet: it cannot tell −0.002 from zero.
- **"Split."** The Preliminary weight is as noisy as the sweep has: 46 to 168 questions move each way under every key.

How large a change can be seen. The baseline intervals above say how well a grade is known in absolute terms, which matters when comparing builds. Two settings on one build are compared paired, which is tighter. Under the five small settings, a question's reciprocal rank changes with a standard deviation of about 0.08 to 0.11. A paired test at 5% with 80% power then detects a change of $2.8 \times \mathrm{sd} / \sqrt{n}$, and needs $7.84 \times (\mathrm{sd} / \delta)^2$ questions to detect a change of δ (Sakai's topic-set-size design, with this project's measured spread; `power.mjs` prints both):

key	questions	smallest detectable change	questions needed for 0.004
ATC (headline)	284	0.018	5,407
ATC repaired	259	0.019	5,926
MeSH	978	0.010	5,592
EPC	469	0.015	6,441
PE	612	0.012	5,382
label, nervous system	350	0.012	3,220
label, every effect	514	0.012	4,473

A change of 0.004 cannot be resolved by any per-question key on this catalogue, however well built. **The keys can arbitrate changes of roughly 0.01 to 0.02 MRR and above, depending on the key. Below that the sweep reports signs, not findings.** These figures are a little lower

than on the 2,452-entry build. That is mostly because the added molecules lowered every reciprocal rank, and with it the spread of the changes. It is not because the keys gained questions: §6 separates the two.

Voorhees 2000 is the reassurance: different assessors' relevance judgements vary widely, yet rank retrieval systems alike. A noisy key still orders formulas whose difference is large. What it cannot do is order formulas whose difference is small, which is where every constant except D15 sits.

---

3. Each key against the literature

Every reference below resolves on Crossref with its title and first author matching (checked 2026-10-16). "Abstract" marks a paper read through its abstract only. "Summary" marks one whose figure comes from a subagent's reading of the full text, which was not checked here.

3.1 ATC, as the headline and repaired

paper	finding	bearing
Winnenburg & Bodenreider 2014, *A framework for assessing the consistency of drug classes across sources*, J Biomed Semantics	Aligning MeSH 2013 with ATC levels 2–4 over 1,685 shared drugs: lexical alignment found 226 equivalent classes and drug-set alignment 223, with only 36 in common; 313 inconsistencies. Same-named classes are often inclusions, not equivalents (summary).	Explains the asymmetric agreement `agreement.mjs` measures (MeSH accepts 79% of ATC's pairs, ATC 19% of MeSH's). Never merge keys by class name.
Liu et al. 2015, *Similarity-based prediction for ATC classification of drugs by integrating multiple data sources*, Bioinformatics	New drugs enter ATC only when someone asks WHO, so coverage is "seriously incomplete" (abstract).	An unclassed molecule is unknown, not unlike: for the filtered reading.
Yıldırım et al. 2007, *Drug–target network*, Nat Biotechnol	Drugs of one ATC type cluster in the drug–target network; follow-on drugs pile onto proteins already targeted (abstract).	For ATC as a target-sharing benchmark, and a reason target sharing partly measures the market.
Mougin, Burgun & Bodenreider 2012, *Comparing drug-class membership in ATC and NDF-RT*	Lexical mapping aligned about half of ATC's terms; mapping cardiovascular classes by hand raised common pairs from 39 to 128 (abstract).	Agreement between ATC and FDA's classes depends on how they are mapped; `agreement.mjs` compares classmate pairs over shared molecules, the safe way.

Measured here. In this catalogue, **95.1% of repaired-ATC classmate pairs share an approved indication** (`confounding.mjs`; random pairs 7.6%). SIMILARITY-LITERATURE §3.5 classed 4 of the 64 level-4 classes as use classes. By what its members are approved for, nearly all of them behave like use classes. The repaired key is therefore as much a "same indication" key as the label key is, and more so. And one class can decide the grade: the five largest classes hold a quarter of the questions. Averaged over classes rather than questions, MRR falls from 0.4341 to 0.3882 (`difficulty.mjs`).

What the literature does not give. No paper tests information-content similarity over the ATC tree (Resnik, Lin; reviewed by Pesquita et al. 2009) as graded relevance against any standard. Replacing the repaired key's flat half credit with such a weight is plausible, untested anywhere, and would change the instrument.

3.2 MeSH pharmacological actions

No published audit of MeSH pharmacological-action indexing was found, for completeness or for errors. Winnenburg & Bodenreider 2014 cite missing action links as one cause of MeSH–ATC misalignment. This project's own audit is the D22 work in MESH-ACTIONS-ROUTE-B.md, and the *Gestormone* error in CLAUDE.md is an NLM indexing error it found.

Measured here. MeSH is the most powerful key: 978 questions, the narrowest interval, and the only key that detected the family weight on the 2,452-entry build. It is powerful because its classes are broad: it accepts 79% of ATC's classmate pairs, and 168 classes hold its 978 questions. It is also the one key the listed entries grew, by 112 scoreable questions (§6). Breadth buys power and costs fidelity, so a change MeSH detects and the narrower keys do not is the one to distrust most. Its MRR falls from 0.5607 to 0.4833 when averaged over classes. Only PE's falls further, from 0.4984 to 0.4040.

3.3 FDA's classes (EPC, MoA, PE)

paper	finding	bearing
Winnenburg, Mortensen & Bodenreider 2015, *Using description logics to evaluate the consistency of drug-class membership relations in NDF-RT*, J Biomed Semantics	The drug-class relations FDA's labels assert and those a reasoner infers from NDF-RT's own drug descriptions "only match in about 50% of the cases" (abstract); 660 of 1,308 drugs have identical class sets (summary).	A true classmate under EPC is often unlabelled as one. Read EPC's filtered grade first.
Pathak & Chute 2010, *Analyzing categorical information in two publicly available drug terminologies: RxNorm and NDF-RT*, JAMIA	About 54% of RxNorm clinical drugs have no NDF-RT counterpart; 42% of MoA classes and 75% of PE classes have no RxNorm drug under them (abstract, as relayed).	Fits the coverage the primer reports (909 molecules whose UNII RxNorm does not know). MoA and PE are sparse by construction.
Nelson et al. 2017, *Interoperability of medication classification systems: lessons learned mapping EPCs to SNOMED CT*	Of 543 EPCs, 284 map to an equivalent SNOMED CT class, 205 are more specific and 54 do not map (abstract, as relayed).	EPC's granularity is its own; no other vocabulary stands in for it.

Measured here. 75.3% of EPC classmate pairs share an approved indication. EPC detects a paired change of about 0.015 (§2). The listed entries added 11 of its 469 questions.

3.4 The label key (OnSIDES)

paper	finding	bearing
Cornelius et al. 2016, *Variation in adverse drug reactions listed in product information for antidepressants and anticonvulsants, between the USA and Europe*, BMJ Open	Twelve CNS brand drugs, each with a US and an EU document from the same company: median 201 against 114 adverse reactions, and "only 29% of ADR terms were reported in both paired documents" (abstract).	The key pools labels across regulators; the regulators largely disagree.

paper	finding	bearing
Duke, Friedlin & Ryan 2011, *A quantitative analysis of adverse events and "overwarning" in drug labeling*, Arch Intern Med	5,602 US labels list a median of 49 and a mean of about 70 distinct adverse events, up to 525 (as relayed; no abstract on PubMed).	Label length is a property of the label as much as of the drug, so profile richness is partly noise. The key's agreement rising with richness (19% to 37%) is what this predicts.
Brouwers et al. 2011, *Network neighbors of drug targets contribute to drug side-effect similarity*, PLoS ONE	Of 1,534 significantly side-effect-similar drug pairs, at least 64% share a target; 5.8% have neighbouring but distinct targets (summary).	For the key: shared targets are the main explanation of shared effects.
Kuhn et al. 2013, *Systematic identification of proteins that elicit drug side effects*, Mol Syst Biol	732 of 1,428 side effects are predicted to be caused chiefly by individual proteins (abstract); indication is not modelled.	For, with the confounder unaddressed.
Duran-Frigola & Aloy 2013, *Analysis of chemical and biological features yields mechanistic insights into drug side effects*, Chem Biol	Most side effects need biological and chemical features together; roughly 38% are explained by biology alone (summary).	A label key will be wrong about many pairs even where it is right in aggregate.
Jamal et al. 2017, *Predicting neurological adverse drug reactions based on biological, chemical and phenotypic properties of drugs*, Sci Rep	Target features predicted neurological adverse reactions least well; phenotypic features, which include indications, best (as relayed).	Indication carries much of the signal in neurological side effects.
Tatonetti et al. 2012, *Data-driven prediction of drug effects and interactions*, Sci Transl Med	Corrects spontaneous reports for covariates including the reason for prescribing, producing OFFSIDES.	The published route to an indication-adjusted effect profile.

OnSIDES's own extraction quality is published for the adverse-reactions section of US labels (F1 0.90; Tanaka et al. 2025). None is published for the UK, EU or Japanese labels, which supply most of the effects for many molecules.

Measured here: the key's reliability (`label_reliability.mjs`, with `label_halves.py`). Each profile is rebuilt under the key's own rules from two disjoint sets of labels: the US labels, and the UK, EU and Japanese ones. Only ingredients with five or more effects in both halves are compared, and neighbours are drawn from that common set.

scope	ingredients	five nearest neighbours shared by the halves	chance
nervous system	152	14.1%	3.3%
every effect	379	17.0%	1.3%

Graded the way the harness grades, on questions both can answer, over the same candidates:

judged by	predicted by	top-1	MRR
non-US half, nervous system (126 questions)	the US half's own similarity list	18.3%	0.3459
	the target ranking	31.0%	0.4796
US half, nervous system (126)	the non-US half's own list	24.6%	0.3981
	the target ranking	34.9%	0.4539
non-US half, every effect (319)	the US half's own list	25.7%	0.3995
	the target ranking	27.6%	0.4124

judged by	predicted by	top-1	MRR
US half, every effect (319)	the non-US half's own list	31.3%	0.4400
	the target ranking	39.5%	0.4892

In all four cases the target ranking predicts a half of the label key better than the other half of the label data does. Two cautions. First, each half is thinner than the pooled key, so the pooled key's reliability is higher than a half's. Second, the halves differ in how much OnSIDES reads from them, so their disagreement mixes the key's noise with a real difference between regulators. The conclusion survives both. The ranking already sits at or above what a second measurement of the key achieves, so the key can rank large changes but has no headroom to rank small ones. Its low absolute grade is the key's ceiling, not the ranking's failure.

Measured here: confounding by indication (`confounding.mjs`). Approved indications are read from ChEMBL's `drug_indication` at phase 4, by exact MeSH id, with forms folded onto parents.

pairs, both with an approved indication	pairs	share one
label neighbours, nervous system	2,042	41.2%
label neighbours, every effect	2,970	46.0%
repaired-ATC classmates	1,281	95.1%
EPC classmates	2,988	75.3%
the ranking's first answer (label-key questions)	264	65.2%
random pairs of labelled molecules	18,418	7.6%

The label key is less a use key than the class keys are. To test whether its agreement with the ranking survives without shared use, the key was rebuilt with every candidate sharing an approved indication with the question set aside. A neighbour must then come from a drug for another disease. The recomputed neighbours match the key's exactly for all 413 questions before anything is set aside.

on the 397 questions with an approved indication	scoreable	top-1	shuffled	MRR
label key as built	340	13.5%	1.6%	0.2503
neighbours from other diseases only	309	1.6%	0.7%	0.0630

Read this carefully, because it does not mean what it first seems to. Drugs with one mechanism are mostly approved for one disease, so setting aside every same-indication candidate also sets aside most same-mechanism ones. The test cannot separate "similar labels because similar mechanism" from "similar labels because similar patients". What it does show is that outside the pairs where use and mechanism coincide, the key and the ranking barely agree. The label key's independence from the score's *inputs* is real. Its independence from *what the drugs are for* is not, and it cannot yet tell mechanism from use any better than ATC can. The literature's remedy is to remove the effects that are the indication's own symptoms (SIDER's approach, Kuhn et al. 2016), not to remove candidates. That keeps the same-mechanism neighbours and takes out the disease.

3.5 The pre-registered pairwise key

paper	finding	bearing
Ryan et al. 2013, *Defining a reference set to support methodological research in drug safety*, Drug Saf	OMOP's 399 test cases, 165 positive and 234 negative controls, each with its level of evidence recorded (abstract, as relayed).	A per-pair evidence level, reported by level.
Coloma et al. 2013, *A reference standard for evaluation of methods for drug safety signal detection using electronic healthcare record databases*, Drug Saf	44 positive and 50 negative controls; two experts verified each independently and a third settled disagreements; only drugs with enough exposure for a method to see were eligible (abstract, as relayed).	Two drafters and an adjudicator; and an eligibility rule like "the pair shares a target", so that a negative can fail.
Hauben et al. 2016, *Evidence of misclassification of drug–event associations classified as gold standard 'negative controls' by OMOP*, Drug Saf	OMOP's rules were asymmetric (a case report made a positive if one textbook cited it, a negative otherwise). Of 233 negatives, 40 (17%) were or may have been misclassified (abstract).	Audit the negatives with a symmetric rule, and budget for errors.
Harpaz et al. 2014, *A time-indexed reference standard of adverse drug reactions*, Sci Data	A reference set dated by label revision, so methods can be tested prospectively.	The same idea as freezing pairs.json before use.
Schuemie et al. 2013, *Interpreting observational studies: why empirical calibration is needed to correct p-values*, Stat Med	Negative controls used as an empirical null: most observational p < 0.05 findings on true nulls disappeared once calibrated (abstract, as relayed).	A change's movement on negatives and opposites is its null: a gain on positives that moves negatives as much is not a gain.
Hripcsak & Wilcox 2002, *Reference standards, judges, and comparison subjects*, JAMIA	Experts serve three roles (making the standard, judging output, being compared), each with its own reliability.	Three drafters can be compared with each other, not only pooled.
Pakhomov et al. 2011, *Towards a framework for developing semantic relatedness reference standards*, J Biomed Inform	13 raters on 101 term pairs; agreement moderate, with systematic rater differences (abstract, as relayed). Their earlier UMNSRS set (AMIA 2010, no DOI) reports drug–drug agreement among four raters of ICC 0.47 (relatedness) and 0.63 (similarity) (summary).	Human agreement on "are these drugs similar" is moderate; no key built by hand should be read as near-perfect.
Walters & Wilder 2023, *Fabrication and errors in the bibliographic citations generated by ChatGPT*, Sci Rep	55% of GPT-3.5's and 18% of GPT-4's citations were fabricated, and many real ones misstated (abstract, as relayed).	The pairs' DOIs were all checked against Crossref by title; whether each cited source *says* what the pair quotes was not checked independently.

Measured here (`difficulty.mjs`):

	directions	clean by construction (no shared target)	of those that could fail, kept below rank 10
negatives	160	96 (60.0%)	62 of 64 (96.9%)
opposites	94	0	89 of 94 (94.7%)

On the 2,452-entry build the last column read 61 of 64 and 85 of 94. The ranking did not get better at these pairs: the added molecules pushed them down (§6). The headline "98.8% of negatives clean" is 64 informative directions in a denominator of 160. They are good news,

but over a quarter of the size the figure suggests. The opposites are the key's hardest and most informative part: every one can fail, and they found the flumazenil gap (SIMILARITY-HARNESS §16).

Of the 116 positives, 64 are repaired-ATC classmates, 81 EPC classmates and 91 MeSH classmates, and **110 are a classmate under at least one**. The positives mostly re-ask questions the class keys already ask; six do not.

3.6 Across keys

- **Circularity, measured.** Grimm et al. 2015 (Hum Mutat) distinguish circularity from shared data and from shared groups (variants of one gene on both sides). Both have analogues here. The first is the class keys restating ChEMBL's curated mechanisms, measured in SIMILARITY-LITERATURE §3.4: 39 of 110 top-1 hits rest on a curated mechanism on both sides. The second is classmates sharing target records. Wallach & Heifets 2018 (J Chem Inf Model) found that a benchmark's redundancy predicted most of the performance reported on it. The redundancy measure worth adding is the share of each key's classmate pairs where both molecules have a curated mechanism at a shared protein, reported per key beside its grade.
- **No latent-class combination.** Dawid & Skene 1979 (Appl Stat) and Hui & Walter 1980 (Biometrics) estimate each judge's error rates without a gold standard. Both assume judges err independently given the truth. ATC, MeSH and EPC share curation, and every key shares indication (§3.4), so the assumption fails here, and a pooled "consensus key" would count one curator's error several times.
- **Hubness grew with the listed entries.** Radovanović et al. 2010 (JMLR) describe nearest-neighbour lists dominated by a few points, and CLAUDE.md records that before normalisation aripiprazole was the top answer for 14 molecules. On the 2,452-entry build the skewness of top-10 appearances was 1.45, the top 1% of molecules filled 5.4% of top-10 places, and the most frequent appeared in 49 lists. On this build they are 2.03, 6.4% and 77, and seven of the ten most frequent are listed entries (`difficulty.mjs`, `growth.mjs`; §6). Skip the listed entries and all three return exactly to the old values. The tested remedies, mutual proximity (Schnitzer et al. 2012) and CSLS (Conneau et al. 2018), would change the score, and nothing here shows the hubs are wrong answers. The figure is worth keeping as a diagnostic after any change to the normalisation or the membership.

---

4. What the measurements say about the keys together

key	smallest detectable change (MRR)	shares with the score	share of classmate pairs with a shared approved indication	own reliability	note
ATC headline	0.018	curated mechanisms, partly	(headline not measured; repaired 95%)	not measurable	mostly structure and use classes; cannot gain a question from new molecules

key	smallest detectable change (MRR)	shares with the score	share of classmate pairs with a shared approved indication	own reliability	note
ATC repaired	0.019	curated mechanisms, partly	95.1%	not measurable	its questions cannot grow either
MeSH	0.010	mechanisms, often	not measured	not measurable	most power, broadest classes; the one key new molecules grew
EPC	0.015	restates mechanism	75.3%	not measurable	about half of memberships unasserted (Winnenburg 2015)
PE	0.012	little	not measured	not measurable	coarse
label	0.012	nothing	41.2%	14% neighbour agreement between regulators	below the ranking's own agreement with it; its answers move with the catalogue
pairwise	not a rate over questions	mechanisms labels state	—	one drafter per area	64 informative negatives, 94 opposites; its top-10 rates move with the catalogue's size

The detectable change is for two settings compared paired on one build, at 5% with 80% power (§2).

Three conclusions follow.

1. **D15 is the only constant every key has tested.** The family weight at 1.0 is detected under MeSH, EPC and PE on this build, under MeSH alone on the last one, and under no key that reads no curation. It crossed the line because the catalogue changed, not because a key got sharper. Every other change in the sweep sits below what any key resolves, so the sweep can neither confirm nor reject it. SIMILARITY-LITERATURE §4.1's rule, "adopt a constant only when judges of different provenance agree", needs a companion: *and only when the change is larger than those judges can resolve*. Without it, the rule would adopt noise whenever two noisy keys happen to agree in sign.
2. **The label key is a check, not an arbiter.** It is the only judge that reads no curation, which makes it the right one to veto a change that only curated judges like. But it is noisier than the ranking, and it rides on indication. Its veto should be read only for changes above its resolution.
3. **The pairwise key's opposites are its most valuable part.** They are reachable, hard, sourced and held out, and they test the direction terms (D15, D19, `OPPOSED_ACTIONS`), which are the part of the score the keys have most firmly shown to matter.

---

5. Brain-specific keys: what the literature offers

The page's question is "may act in similar ways in the brain". None of the keys above measures the brain directly. The literature has five candidates; none is ready at catalogue scale.

candidate	evidence that target profiles predict it	data	verdict
Subjective-effect reports of psychedelics	Zamberlan et al. 2018: the similarity of binding-affinity profiles of substituted phenethylamines and tryptamines correlates with the semantic similarity of their Erowid reports (abstract; R = 0.60 on one affinity set and 0.48 on another, summary). Ballentine et al. 2022 map words in 6,850 reports on 27 drugs to receptor expression.	Erowid reports (terms of use unchecked); 18–27 drugs	the only direct test of target profile against brain effect in this survey; a pilot for one class, not a key
Receptor affinity against clinical effects within a class	Kroeze et al. 2003: H1 affinity predicts short-term weight gain across 17 antipsychotics (as relayed).	trial and review tables	within-class checks, a few dozen drugs
Pharmaco-EEG fingerprints	Dimpfel 2003 (Eur J Med Res, PMID 12844474): rat EEG classifies drugs into eight CNS categories (abstract, as relayed).	no open per-drug table found	not buildable until a dataset is found
Zebrafish behavioural profiles	Rihel 2010, Kokel 2010, Bruni 2016 (in SIMILARITY-LITERATURE §4.2)	supplementary tables, joined by name	buildable with care; name joins break invariant 15 unless each compound is mapped by identifier
Drug discrimination	the classic "feels alike" measure	no machine-readable database of generalisation results found; review literature only	a hand-curated, sourced pilot in the pairwise key's style
Mouse behavioural screening (SmartCube)	Roberds et al. 2011, Alexandrov et al. 2015	proprietary reference database	not obtainable

6. What adding molecules does to the keys (`growth.mjs`)

Between the two versions of this review the catalogue grew from 2,452 to 3,291 entries. Nearly all of that is D25–D27, which entered 841 molecules because Wikipedia, Wikidata, TripSit, PsychonautWiki or the sci-wiki lists them. They are mostly research compounds and recreational substances: 35 of the 841 are approved in ChEMBL's record, and 37 carry any ATC code. That makes them a natural experiment in what a larger catalogue does to keys built from labels.

`node analysis/answer_keys/growth.mjs` reads the 3,291-entry build twice: as built, and with the listed entries taken out of every list (`skippingListed`, the reading the floor test already uses). The second reading is the same ranking among the same old molecules, so the difference between the two is what the additions did. The 2,452-entry build is quoted beside them and not subtracted, because more than the listing changed between the builds (§8).

6.1 What the 841 did

Questions. A molecule becomes a question only when a key labels it and a classmate shares a target with it:

key	questions, 2,452 build	questions, 3,291 build	of them listed	MRR, 2,452 build	MRR as built	MRR of the old questions, listed skipped	smallest detectable change, 2,452 → 3,291
ATC (headline)	284	284	0	0.4607	0.4341	0.4644 (284)	0.019 → 0.018
ATC repaired	259	259	0	0.4958	0.4640	0.4993 (259)	0.021 → 0.019
MeSH	865	978	112	0.5945	0.5607	0.5931 (864)	0.011 → 0.010
EPC	455	469	11	0.5946	0.5650	0.5983 (455)	0.015 → 0.015
PE	598	612	15	0.5154	0.4984	0.5150 (597)	0.013 → 0.012
label, nervous system	346	350	4	0.2620	0.2512	0.2639 (346)	0.014 → 0.012
label, every effect	507	514	7	0.3180	0.3056	0.3179 (506)	0.013 → 0.012

Six things follow.

1. **Questions come from labels, not from molecules.** The listed entries are a quarter of the catalogue and 3% of the label key's questions. Only MeSH, whose supplementary records reach research compounds, gained a real number, 112. Neither ATC key can gain any. An ATC N code admits a molecule by itself, so all 544 parents in ChEMBL 37 with an N code are already in, and both ATC keys ask only about N-coded molecules. The repaired key takes its peers from every ATC level, so adding drugs with other ATC codes could make more of its 353 labelled questions scoreable. It can never make more than 353.

2. **Unlabelled additions lower every grade as built, and the ranking is no worse for it.** A listed entry ranked above a classmate is a miss under every class key however apt it is (SIMILARITY-HARNESS §11 and §18 say the same of D21 and D25). Read with the listed entries skipped, the old questions grade within 0.004 of the 2,452-entry build under every key. What gap remains is what else changed between the builds (§8). So a grade compared across two builds as built mixes the catalogue's growth with the ranking's quality. The filtered reading, or `--skip-listed` over the old questions, does not. The new questions themselves are harder: MeSH's 971 with listed entries skipped grade 0.5766, its 864 old ones 0.5931.

3. **The resolution barely moved, and mostly for the wrong reason.** As built, the smallest detectable changes are a little smaller. But that is because the added molecules push every classmate down, which lowers every reciprocal rank and with it the spread of per-question changes (under ATC, sd 0.115 with the listed entries skipped, 0.105 as built). The effects being measured shrink in the same proportion (D15 under ATC: −0.0295 skipped, −0.0270 as built). Relative to the grade, ATC resolves about 4% of its MRR on both builds. Only the new questions are a real gain: MeSH's 112, which moves its threshold from 0.011 to 0.010.

4. **The pairwise key's top-10 rates depend on the catalogue's size.** Its pairs are fixed, but the listed entries lengthen every list: for a molecule not listed, the median list has 179 entries as built and 164 with them skipped.

	2,452 build	3,291 as built	3,291, listed skipped
positives within the top 10	75.4%	69.8%	75.0%
negatives kept out of it	98.1%	98.8%	98.8%
opposites kept out of it	90.4%	94.7%	92.6%
a positive above a negative or opposite	96.0%	96.0%	—

   The positive rate falls and the opposite rate rises without the ranking changing its order among the pairs. A catalogue change can therefore look like a trade between the two, or, summed, like nothing. The ordering figure, which compares partners within one list, is the one that does not move with the catalogue's size. It should be read first (§7, step 2).

5. **The label key's answers move with the catalogue.** Its classmates are each question's five nearest label neighbours *among the catalogue's molecules*, so a newly entered molecule with a label can become one. On the two builds, 30 of the 409 shared nervous-system questions have a different answer set: 31 of 2,176 neighbours were replaced, 14 of them by molecules entered since. Under every effect, 67 of 609 changed. Version 1.0.0 froze the rules but not the answers, so a v1.0.0 grade on two builds is two slightly different keys (§7, step 3).

6. **The new molecules became the ranking's hubs, mostly in each other's lists.** Skewness of top-10 appearances went from 1.45 to 2.03, and the most frequent molecule from 49 lists to 77. Seven of the ten most frequent are listed entries: cannabinoid ligands (AM-1241, CB-13), a tryptamine (5-MeO-αMT), a cathinone, muscarine, homomazindol and 7-hydroxymitragynine. They fill 58.3% of the top-10 places in listed entries' lists and 15.2% in everyone else's. With them skipped, all three hubness figures return exactly to the 2,452-entry build's. The entries come in families (synthetic cannabinoids, cathinones, tryptamines), and these figures say a family mostly fills its own members' lists. Why the few that reach into others' lists do so was not examined.

One sweep result moved with the catalogue. The family weight at 1.0 was detected under MeSH alone on the 2,452-entry build. On this build MeSH, EPC and PE detect it, and with the listed entries skipped, MeSH and PE detect it and EPC does not (§2). None of the keys is sharper: the lists they grade changed, and why EPC's moved this way was not examined. This is why a constant should be compared between two settings on one build, never across a catalogue change.

6.2 What adding more would do

Under the paired test of §2, the smallest detectable change is 2.8 × sd / √n for n questions, so halving it takes four times the questions. Detecting the sweep's +0.004 would take 3,200 to 6,400 questions depending on the key, against 284 to 978 now. Each way the catalogue could grow gives this much:

route	how many	what it gives the keys	best smallest detectable change
more molecules like D25–D27's	the four open lists are spent: D27 entered what they name that ChEMBL 37 holds, except the placements its review refused (COHORT-2, D27)	13% become MeSH questions, at most 2% anything else	MeSH 0.004 only after about 34,600 more such molecules
every approved drug in ChEMBL 37 not yet in	about 1,745 (2,996 approved parents; 1,251 in)	at the yields of the approved drugs already in: per molecule, 0.59 MeSH questions, 0.48 PE, 0.41 label (every effect), 0.37 EPC, 0.28 label (nervous system), 0 ATC	MeSH about 0.007; PE and both label readings about 0.008; EPC about 0.010
any route, for the ATC keys	—	none: their questions are the 544 N-coded parents, already in	stays at 0.018–0.019

Two warnings on the second row. First, those drugs are approved almost entirely for diseases outside the adopted scope; were their indications in scope, the indication route would already have admitted them. Their questions would be about antibiotics, cancer drugs and antihypertensives, so the keys would be grading the score on something other than the page's question. Second, admitting them is a membership change, which is the user's decision (CLAUDE.md, *Not done*). It is costed here, not proposed.

So the catalogue cannot grow its way to resolving the sweep's small changes. The bound is how many labelled drugs exist, not how many molecules ChEMBL holds. What adding molecules does reliably is the opposite of what a sweep needs: it lowers grades as built, crowds the pairwise key's top 10, changes the label key's answers and adds hubs, all without the ranking changing. Power has to come from elsewhere. One source is questions that carry more information each, such as pairwise negatives that can fail (§7, step 2). The other is not sweeping for changes too small to see. Growing the catalogue is worth doing for the page, as long as every comparison is made on one build or with the new entries skipped.

---

7. Proposed order

Each step is its own change and adopts nothing about the score.

Step 1: fix the decision rule before the next constant moves (no new data).

- Report every sweep cell with its paired interval, question-level and class-resampled, and the Holm-adjusted test across the cells actually run. `power.mjs` does this for six settings. Folded into `sweep.mjs`, it would do it for every grid point.
- State each key's resolution beside its grade, as Status states coverage.

- Write the adoption rule down, versioned like a key. A suggested form: a change must be detected (Holm) under at least two keys of different provenance, one of them reading no curation; must not worsen the pairwise key's negatives or opposites; and is confirmed once on the held-out pairwise key after it is chosen, never before. Under that rule, of the six settings in §2 only D15 would qualify, and it is already adopted. The family weight is detected under three curated keys on this build, but under no key that reads no curation, so it would not.
- Report MRR averaged over classes beside MRR averaged over questions.
- Compare two builds only with the entries a key cannot label skipped in each list (`--skip-listed`, as the floor test already grades), and state the catalogue's size beside every grade. Compare two settings only on one build (§6).

Step 2: pairwise key v2. Keep v1 frozen; add, never edit.

- Draw new negatives from pairs that **share at least one target** but differ in primary mechanism (an antipsychotic and an antihistamine at H1; a tricyclic and an anticholinergic at muscarinic receptors). These can fail, and they are what the score most needs to get right. Report negative rates over reachable directions beside the current figure.
- Have a second drafter, blind to the first and from a different model family, label every pair, with an adjudicator. Report agreement by kind (Coloma 2013; Pakhomov 2011).
- Check that each cited source says what the pair quotes, not only that the DOI resolves. Record each pair's evidence level (label, trial, review, case report), and grade by level (Ryan 2013).
- Audit v1's negatives with one symmetric rule for what disqualifies a negative, and budget for Hauben's 17%.
- Grow the opposites, the part that has found a real gap.
- Report a reading that does not depend on the catalogue's size beside the top-10 rates: the share of lists ranking a positive above a negative or opposite, which v1 already reports (96.0% on both builds), or each partner's rank as a share of its list. v1's frozen rules are not changed for this; the new reading sits beside them (§6).

Step 3: label key v2. Under SIMILARITY-LITERATURE §4.1 a new key needs a new version and a changelog entry; v1 stays.

- Require an effect to appear in labels from **two regulators**, or report a US-only and a non-US reading beside the pooled one (Cornelius 2016). `label_reliability.mjs` measures each version's reliability, and a v2 that does not raise it is not better.
- Remove from each profile the effects that are its approved indications' own symptoms, through an identifier mapping from the indication's MeSH heading to MedDRA terms, never by name. Then rerun `confounding.mjs`: the test is whether the "other diseases only" row rises above its shuffle.
- Consider OFFSIDES (Tatonetti 2012) as an indication-adjusted second source.
- Freeze each version's answers, not only its rules. Write each question's neighbours to a file when the version is cut, so that a catalogue change cannot change the key under the same version number (§6).

Step 4: graded relevance by information content, as one alternative reading of the repaired ATC key and the MeSH key, labelled untested (Pesquita 2009).

Step 5: a brain-specific pilot. Two candidates: a sourced drug-discrimination subset added to the pairwise key's opposites and positives, or the psychedelic subjective-effect set as a within-class check. Neither will be large enough to arbitrate a constant. Both would test the page's own question.

What not to do. Do not add more class keys to raise power: they share curation and indication, and agreement among them is not independent evidence. Do not pool the keys into a consensus by latent-class methods: their errors are correlated. And do not keep sweeping for changes under about 0.01 MRR: on this catalogue no key can see them, and a constant chosen there is chosen by noise. Do not expect a larger catalogue to change that (§6).

---

8. How much to trust this

- **Citations.** Three literature sweeps were run by subagents in parallel (class keys, phenotype keys, evaluation method). Every DOI cited here was resolved on Crossref on 2026-10-16, and its title and first author matched. The abstracts behind three load-bearing figures, Cornelius 2016's 29%, Hauben 2016's 17% and Winnenburg 2015's 50%, were read here. Rows marked *summary* rest on a subagent's reading of a full text, and rows marked *as relayed* on a subagent's reading of an abstract, neither checked here. Those figures should be checked before anything depends on them. Pakhomov et al. 2010 (UMNSRS) and Dimpfel 2003 have no DOI.
- **Measurements.** Every figure in §2–§4 comes from `power.mjs`, `label_reliability.mjs` (with `label_halves.py`), `confounding.mjs`, `difficulty.mjs` and `agreement.mjs`, and §6's from `growth.mjs` and `power.mjs --skip-listed`. All were run on 2026-10-31 against the build of 2026-10-16, with the two answer-key backfills run on it that day. They read the payload, `db/cohort1.db`, the cached OnSIDES zip and, for indications, ChEMBL 37 at `$CHEMBL_SQLITE_PATH`. They write nothing. The bootstraps are seeded, so a rerun prints the same intervals. The 2,452-entry figures in §6 are the first version's, from the 2026-10-13 build. One comparison in §6, the label key's answers on the two builds, was made once against a saved copy of that build, and the repository cannot rerun it.
- **The two builds differ by more than the listing.** Between them came D23 and D24 (an Approved taken back where only unapproved labels support it), GtoPdb 2026.3.1, and the Wikipedia and viewer work. None of these changes the score's formula, but the GtoPdb release changes some claims. So the clean comparison in §6 is the same build read twice, as built and with the listed entries skipped, and the old build is quoted beside it, not subtracted from it.
- **The detectable changes and sample sizes** (§2) use each key's median standard deviation of per-question changes over the five small settings, and the normal approximation. They are orders of magnitude, not a design; a larger change moves more questions further, so its spread is larger too.
- **The indication test** compares exact MeSH ids. "Depressive Disorder" and "Depressive Disorder, Major" count as different, which undercounts sharing, so the true share for every row in §3.4 is somewhat higher.
- **The projections in §6** carry the yield per molecule of one kind of addition over to another, and assume the spread of per-question changes stays where it is. They give orders of magnitude.

- **Not measured.** The MeSH key's share of same-indication pairs; the redundancy measure proposed in §3.6; the agreement between the three pairwise drafters, which cannot be measured since each drafted a different area; and OnSIDES's extraction quality outside the US.